1. **Identify the dimensions from each dimension table**

bed_type.csv (Dimension Table)

Dimensions:

- bed_id (Primary Key - dimension)
- bed_code (dimension)
- bed_desc (dimension)

-> bed_type is a dimension table because bed_id, bed_code and bed_desc are not numerical measurement. We can not measure or sum up bed_id even though it is numeric because bed_id is an unique identifier for identifying or categorizing bed types

bed_id	bed_code	bed_desc
1	BU	Burn
2	CC	CCU
3	DE	Detox ICU

business.csv (Dimension Table)

Dimensions:

- ims_org_id (Primary Key - dimension)
- business_name (dimension)
- bed_cluster_id (dimension)

Facts:

- ttl_license_beds (fact - numerical measure)
- ttl_census_beds (fact - numerical measure)
- ttl_staffed_beds (fact - numerical measure)

-> Even though business has both dimensions and facts, business is still a dimension table in the context of the STAR schema with bed_fact because business provides descriptive information about hospitals and the primary key (ims_org_id) is used as a foreign key in bed_fact.

ims_org_id	business_name	ttl_license_beds	ttl_census_beds	ttl_staffed_beds	bed_cluster_id
INS00000519	Abington Memorial Hospital	671	466	671	5
INS00000520	Lower Oconee Community Hospital, Inc	25	14	25	1
INS00000521	Stewart Webster Hospital, Inc	25	16	25	1
INS00000522	Glens Falls Hospital	410	300	315	3
INS00000523	Washington Adventist Hospital	288	237	288	3
INS00000524	Shady Grove Adventist Hospital	336	300	336	3
INS00000525	Hackettstown Regional Medical Center	111	60	111	1
INS00000526	Castle Medical Center	160	105	160	2
INS00000527	Feather River Hospital	101	70	80	1

2. **Identify the Facts variables from the single Fact Table**

bed_fact.csv (Fact Table)

Dimensions:

- ims_org_id (Foreign Key - dimension linking to business table)
- bed_id (Foreign Key - dimension linking to bed_type table)

Facts:

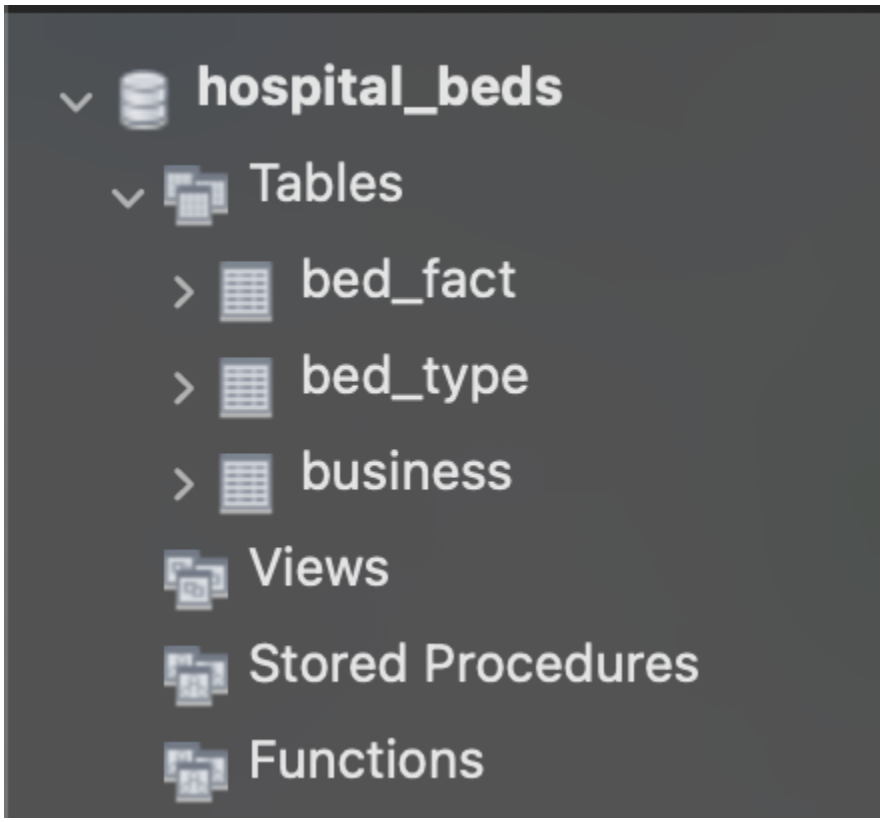
- license_beds (fact - numerical measure)
- census_beds (fact - numerical measure)
- staffed_beds (fact - numerical measure)

-> bed_fact is a fact table because:

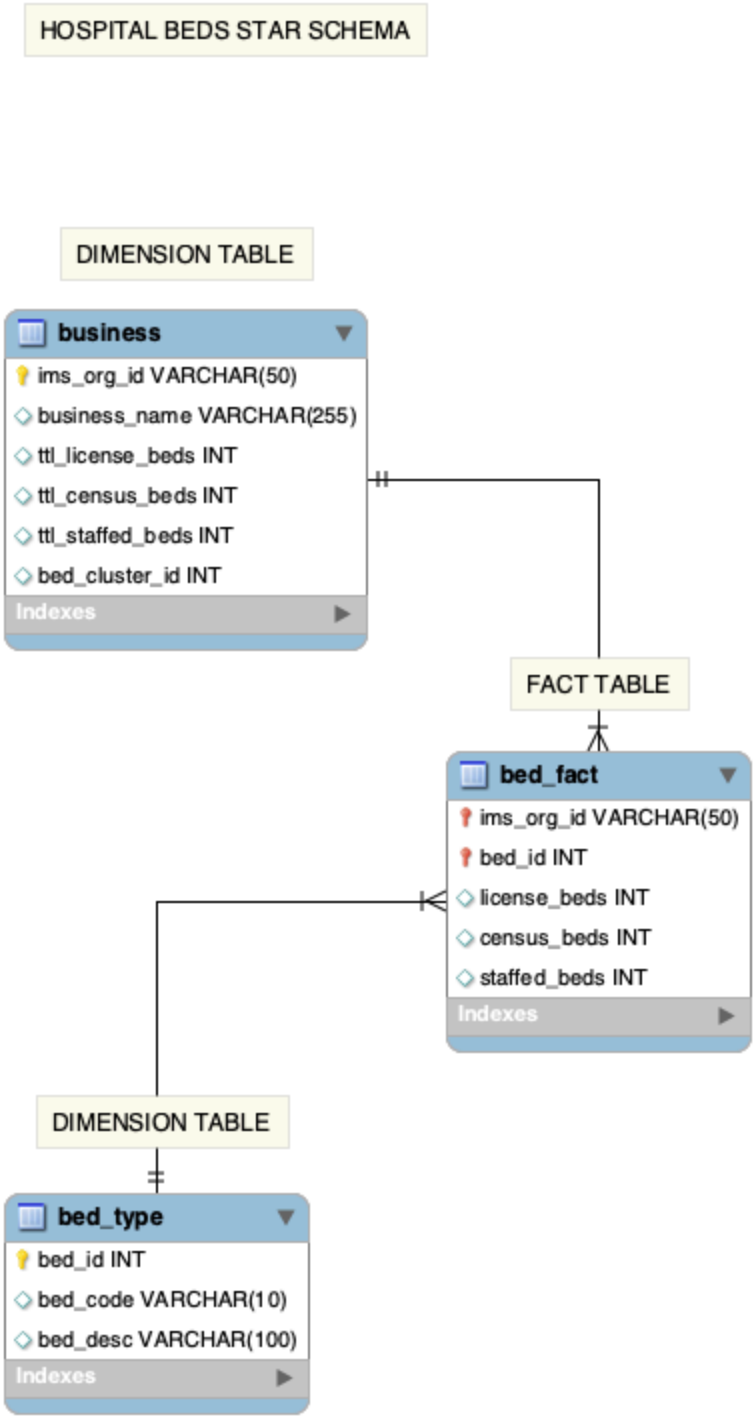
- ims_org_id: is dimension and is used for identifying which hospitals
- bed_id: is dimension and is used for identifying which type of beds
- license_beds: is fact because we can measure how many licensed beds this hospital has
- census_beds: is fact because we can measure how many census beds this hospital has
- staffed_beds: is fact because we can measure how many beds for staffs this hospital has

ims_org_id	bed_id	license_beds	census_beds	staffed_beds
INS00000519	2	10	7	10
INS00000519	4	38	26	38
INS00000519	5	566	394	566
INS00000519	6	32	22	32
INS00000519	15	25	17	25
INS00000519	18	671	466	671

3. **Create a new schema in MySQL Workbench and add the downloaded .csv files as tables.**



4. **Use ALTER TABLE SQL commands OR use MySQL Workbench to specify the relationships between tables.**
5. **Use "Reverse Engineering Wizard" to create a ERD of a Star Schema using MySQL Workbench**



6.

a. Analysis for Leadership

```
36
37    -- Top 10 Hospitals by ICU/SICU License Beds
38
```

100% 11:48

Result Grid Filter Rows: Search Export:

hospital_name	total_license_be...
Phoenix Childrens Hospital	247
University of Maryland Medical Center	220
UC Health University Hospital	218
Wesley Medical Center, LLC	214
Vidant Medical Center	204
Rady Childrens Hospital and Health Center	200
Dallas County Hospital Association	195
Saint Lukes Episcopal Hospital Texas Medical C...	178
The Methodist Hospital	170
Emory University Hospital	169

```
50    -- Top 10 Hospitals by ICU/SICU Census Beds
51
```

100% 10:61

Result Grid Filter Rows: Search Export:

hospital_name	total_census_be...
Shands Hospital at the University of Florida	167
Dallas County Hospital Association	145
Mercy Medical Center Saint Louis	142
Los Angeles County University of Southern Calif...	139
The Methodist Hospital	138
University of Minnesota Medical Center Fairview	129
University of Maryland Medical Center	127
Brigham and Womens Hospital	124
Vidant Medical Center	123
Ronald Reagan University of California Los Ang...	122

```
61    -- Top 10 Hospitals by ICU/SICU Staffed Beds
62
```

100% 10:71

Result Grid Filter Rows: Search Export:

hospital_name	total_staffed_b...
Vidant Medical Center	203
Rady Childrens Hospital and Health Center	200
University of Maryland Medical Center	171
Emory University Hospital	169
Shands Hospital at the University of Florida	167
Mercy Medical Center Saint Louis	163
Wesley Medical Center, LLC	162
Phoenix Childrens Hospital	159
Grady Memorial Hospital	154
UC Health University Hospital	151

b. Interpretation of Findings

- License Beds: Phoenix Childrens Hospital and University of Maryland Medical Center lead here with 247 and 220 relatively. These hospitals have state approval for the most beds.
- Census Beds: Shands Hospital at the University of Florida and Dallas County Hospital Association are at the top with 167 and 145 relatively, using the most ICU beds at a specific time, which means they are also having high patient counts.
- Staffed Beds: Vidant Medical Center and Rady Childrens Hospital and Health Center lead here with 203 and 200 beds for staffs.

Key Insight:

The most important finding is that the same few hospitals keep appearing across different lists. This means they have high capacity, can staff it, and are using it. These hospitals are:

- University of Maryland Medical Center
- Shands Hospital at the University of Florida
- Vidant Medical Center

7.

a. Drill down investigation

```
-- Step 7a: Drill Down - Hospitals with BOTH ICU AND SICU
-- Top 10 with Both ICU & SICU - License Beds
```

100% 10:91

Result Grid Filter Rows: Search Export: Fetch rows:

hospital_name	total_license_be...
University of Maryland Medical Center	220
UC Health University Hospital	218
Shands Hospital at the University of Florida	167
MCGHealth, Inc	155
Grady Memorial Hospital	154
Jackson Memorial Hospital	151
University Hospital in Bexar County	144
University of Minnesota Medical Center Fairview	144
Carolinas Medical Center	137
Yale New Haven Hospital	136

```
-- Top 10 with Both ICU & SICU - Census Beds
SELECT
```

100% 45:93

Result Grid Filter Rows: Search Export:

hospital_name	total_census_be...
Shands Hospital at the University of Florida	167
University of Minnesota Medical Center Fairview	129
University of Maryland Medical Center	127
Jackson Memorial Hospital	117
UC Health University Hospital	110
Carolinas Medical Center	106
Cedars Sinai Health System	92
Duke University Health System	91
University Hospital in Bexar County	91
Grady Memorial Hospital	88

```
-- Top 10 with Both ICU & SICU - Staffed Beds
SELECT
```

100% 8:112

Result Grid Filter Rows: Search Export:

hospital_name	total_staffed_b...
University of Maryland Medical Center	171
Shands Hospital at the University of Florida	167
Grady Memorial Hospital	154
UC Health University Hospital	151
University of Minnesota Medical Center Fairview	144
Carolinas Medical Center	137
Chattanooga Hamilton County Hospital Authority	134
Saint Josephs Hospital and Medical Center	134
Jackson Memorial Hospital	128
Sunrise Hospital and Medical Center, LLC	125

b. Final recommendation

Interpretation:

- License Beds: University of Maryland Medical Center and UC Health University Hospital lead here with 220 and 218 relatively.
- Census Beds: Shands Hospital at the University of Florida and University of Minnesota Medical Center Fairview are at the top with 167 and 129 relatively.
- Staffed Beds: University of Maryland Medical Center and Shands Hospital at the University of Florida lead here with 171 and 167 beds for staffs.

Key Insight:

After two analysis and investigation, **University of Maryland Medical Center and Shands Hospital at the University of Florida** are consistently the top performers, especially when focusing on hospitals that have both ICU and SICU units.

Final Recommendation:

I recommend the two following hospitals for the pilot program: **University of Maryland Medical Center** and **Shands Hospital at the University of Florida** because they have high bed capacity and especially for both ICU/SICU care. This makes them ideal for testing new hired nurses for critical care patients. These two hospitals have the scale, the staffs and the patient counts to make this intervention cost-effective.